MATH 350-2 Advanced Calculus

W.R. Casper

Department of Mathematics
California State University Fullerton

September 7, 2025

Outline

Outline

1. Real Analysis Lecture 4
   - Functions
   - More with cardinality

Functions

A **function** from A to B is a relation $\mathcal{R}$ from A to B with the property

Functions

A **function** from A to B is a relation $\mathcal{R}$ from A to B with the property

$$a\mathcal{R}b \text{ and } a\mathcal{R}c \quad \text{implies} \quad b = c.$$

Functions

A **function** from A to B is a relation $\mathcal{R}$ from A to B with the property

$$a\mathcal{R}b \text{ and } a\mathcal{R}c \quad \text{implies} \quad b = c.$$

If a relation $\mathcal{R}$ is a function, we usually use a symbol like f.

Functions

A **function** from A to B is a relation $\mathcal{R}$ from A to B with the property

$$a\mathcal{R}b \text{ and } a\mathcal{R}c \quad \text{implies} \quad b = c.$$

If a relation $\mathcal{R}$ is a function, we usually use a symbol like f.

NOTATION: $f : A \to B$ means f is a function from A to B

NOTATION: $f(a) = b$ means $(a, b) \in f$.

Functions

A **function** from A to B is a relation $\mathcal{R}$ from A to B with the property

$$a\mathcal{R}b \text{ and } a\mathcal{R}c \quad \text{implies} \quad b = c.$$

If a relation $\mathcal{R}$ is a function, we usually use a symbol like f.

NOTATION: $f : A \to B$ means f is a function from A to B

NOTATION: $f(a) = b$ means $(a, b) \in f$.

The set

$$\text{img}(f) = \{f(a) : a \in A\}$$

is called the **range** or **image** of f

Challenge!

Problem

Is the relation on $\mathbb{R}$ defined by

$$\{(x, y) : x^2 + y^2 = 1\}$$

Challenge!

Problem

Is the relation on $\mathbb{R}$ defined by

$$\{(x, y) : x^2 + y^2 = 1\}$$

(a) reflexive?

Challenge!

Problem

Is the relation on $\mathbb{R}$ defined by

$$\{(x, y) : x^2 + y^2 = 1\}$$

(a) reflexive?
(b) symmetric?

Challenge!

Problem

Is the relation on $\mathbb{R}$ defined by

$$\{(x, y) : x^2 + y^2 = 1\}$$

(a) reflexive?
(b) symmetric?
(c) transitive?

Challenge!

Problem

Is the relation on $\mathbb{R}$ defined by

$$\{(x, y) : x^2 + y^2 = 1\}$$

(a) reflexive?
(b) symmetric?
(c) transitive?
(d) a function?

Properties of functions

A function f from A to B is called **one-to-one** or **injective** if

Properties of functions

A function f from A to B is called **one-to-one** or **injective** if

$$f(x) = f(y) \quad \text{implies} \quad x = y \;\text{ for all } x, y \in A.$$

Properties of functions

A function f from A to B is called **one-to-one** or **injective** if

$$f(x) = f(y) \quad \text{implies} \quad x = y \ \text{ for all } x, y \in A.$$

It is called **onto** or **surjective** if $\mathrm{img}(f) = B$, or equivalently

$$\text{for all } b \in B \text{ there exists } a \in A \text{ with } f(a) = b.$$

Properties of functions

A function f from A to B is called **one-to-one** or **injective** if

$$f(x) = f(y) \quad \text{implies} \quad x = y \ \text{ for all } x, y \in A.$$

It is called **onto** or **surjective** if $\operatorname{img}(f) = B$, or equivalently

$$\text{for all } b \in B \text{ there exists } a \in A \text{ with } f(a) = b.$$

If it satisfies both properties, it is called **bijective.**

Inverses

The **converse** of a relation $\mathcal{R}$ from A to B is the relation

Inverses

The **converse** of a relation $\mathcal{R}$ from A to B is the relation

$$\check{\mathcal{R}} = \{(b, a) \in B \times A : a\mathcal{R}b\}.$$

Inverses

The **converse** of a relation $\mathcal{R}$ from A to B is the relation

$$\check{\mathcal{R}} = \{(b, a) \in B \times A : a\mathcal{R}b\}.$$

If f is function, then $\check{f}$ may or not be a function.

Inverses

The **converse** of a relation $\mathcal{R}$ from A to B is the relation

$$\check{\mathcal{R}} = \{(b, a) \in B \times A : a\mathcal{R}b\}.$$

If f is function, then $\check{f}$ may or not be a function.
If it is, we call it the **inverse** of f.

Compositions

The composition of $f : A \to B$ and $g : B \to C$ is the function

Compositions

The composition of $f : A \to B$ and $g : B \to C$ is the function

$$g \circ f : A \to C$$

Compositions

The composition of $f : A \to B$ and $g : B \to C$ is the function

$$g \circ f : A \to C$$

with domain A and codomain C defined by

Compositions

The composition of $f : A \to B$ and $g : B \to C$ is the function

$$g \circ f : A \to C$$

with domain A and codomain C defined by

$$(g \circ f)(x) = g(f(x)).$$

Sequences

A **finite sequence** is a function $f : \{1, 2, \ldots, n\} \to \mathbb{R}$.

Sequences

A **finite sequence** is a function $f : \{1, 2, \dots, n\} \to \mathbb{R}$.
An **infinite sequence** is a function $f : \mathbb{Z}_+ \to \mathbb{R}$.

Sequences

A **finite sequence** is a function $f : \{1, 2, \ldots, n\} \to \mathbb{R}$.
An **infinite sequence** is a function $f : \mathbb{Z}_+ \to \mathbb{R}$.

NOTATION: f_n indicates the value $f(n)$

Sequences

A **finite sequence** is a function $f : \{1, 2, \ldots, n\} \to \mathbb{R}$.
An **infinite sequence** is a function $f : \mathbb{Z}_+ \to \mathbb{R}$.

NOTATION: f_n indicates the value $f(n)$

NOTATION: $\{f_n\}$ is another way of writing the function f

Sequences

A **finite sequence** is a function $f : \{1, 2, \dots, n\} \to \mathbb{R}$.
An **infinite sequence** is a function $f : \mathbb{Z}_+ \to \mathbb{R}$.

NOTATION: f_n indicates the value $f(n)$

NOTATION: $\{f_n\}$ is another way of writing the function f

If $k : \mathbb{Z}_+ \to \mathbb{Z}_+$ is a function which is **strictly increasing**, meaning

$$m < n \quad \Rightarrow \quad k(m) < k(n),$$

Sequences

A **finite sequence** is a function $f : \{1, 2, \ldots, n\} \to \mathbb{R}$.
An **infinite sequence** is a function $f : \mathbb{Z}_+ \to \mathbb{R}$.

NOTATION: f_n indicates the value $f(n)$

NOTATION: $\{f_n\}$ is another way of writing the function f

If $k : \mathbb{Z}_+ \to \mathbb{Z}_+$ is a function which is **strictly increasing**, meaning

$$m < n \quad \Rightarrow \quad k(m) < k(n),$$

then the composition $f \circ k : \mathbb{Z}_+ \to \mathbb{R}$ forms a sequence called a **subsequence** of f.

Sequences

A **finite sequence** is a function $f : \{1, 2, \ldots, n\} \to \mathbb{R}$.
An **infinite sequence** is a function $f : \mathbb{Z}_+ \to \mathbb{R}$.

NOTATION: f_n indicates the value $f(n)$

NOTATION: $\{f_n\}$ is another way of writing the function f

If $k : \mathbb{Z}_+ \to \mathbb{Z}_+$ is a function which is **strictly increasing**, meaning

$$m < n \quad \Rightarrow \quad k(m) < k(n),$$

then the composition $f \circ k : \mathbb{Z}_+ \to \mathbb{R}$ forms a sequence called a **subsequence** of f.

NOTATION: $\{f_{k(n)}\}$ or $\{f_{k_n}\}$ both really mean $f \circ k$

Cantor's Paradise

Theorem (Cantor-Schroeder-Bernstein Theorem)

The following are equivalent

(i) $|A| \leq |B|$ *and* $|B| \leq |A|$
(ii) $|A| \geq |B|$ *and* $|B| \geq |A|$
(iii) $|A| \leq |B|$ *and* $|A| \geq |B|$
(iv) $|A| = |B|$

Challenge

Problem

Show that $\mathbb{Z}_+$ and $\mathbb{Z}_+ \times \mathbb{Z}_+$ have the same cardinality.

Challenge

Problem

Show that $\mathbb{Z}_+$ and $\mathbb{Z}_+ \times \mathbb{Z}_+$ have the same cardinality.

Hint: consider $f : \mathbb{Z}_+ \times \mathbb{Z}_+ \to \mathbb{Z}_+$, $f(m, n) = 2^m 3^n$

Solution

Challenge

Problem

Show that $\mathbb{Z}_+$ and $\mathbb{Z}_+ \times \mathbb{Z}_+$ have the same cardinality.

Hint: consider $f : \mathbb{Z}_+ \times \mathbb{Z}_+ \to \mathbb{Z}_+$, $f(m, n) = 2^m 3^n$

Solution

$2^a 3^b = 2^c 3^d$ if and only if $a = c$ and $b = d$ (Fund. Thm. Arith.)

Challenge

Problem

Show that $\mathbb{Z}_+$ and $\mathbb{Z}_+ \times \mathbb{Z}_+$ have the same cardinality.

Hint: consider $f : \mathbb{Z}_+ \times \mathbb{Z}_+ \to \mathbb{Z}_+$, $f(m, n) = 2^m 3^n$

Solution

$2^a 3^b = 2^c 3^d$ if and only if $a = c$ and $b = d$ (Fund. Thm. Arith.)
$f(m, n) = 2^m 3^n$ is injective

Challenge

Problem

Show that $\mathbb{Z}_+$ and $\mathbb{Z}_+ \times \mathbb{Z}_+$ have the same cardinality.

Hint: consider $f : \mathbb{Z}_+ \times \mathbb{Z}_+ \to \mathbb{Z}_+$, $f(m,n) = 2^m 3^n$

Solution

$2^a 3^b = 2^c 3^d$ if and only if $a = c$ and $b = d$ (Fund. Thm. Arith.)
$f(m,n) = 2^m 3^n$ is injective
$|\mathbb{Z}_+ \times \mathbb{Z}_+| \leq |\mathbb{Z}_+|$

Challenge

Problem

Show that $\mathbb{Z}_+$ and $\mathbb{Z}_+ \times \mathbb{Z}_+$ have the same cardinality.

Hint: consider $f : \mathbb{Z}_+ \times \mathbb{Z}_+ \to \mathbb{Z}_+$, $f(m, n) = 2^m 3^n$

Solution

$2^a 3^b = 2^c 3^d$ if and only if $a = c$ and $b = d$ (Fund. Thm. Arith.)
$f(m, n) = 2^m 3^n$ is injective
$|\mathbb{Z}_+ \times \mathbb{Z}_+| \leq |\mathbb{Z}_+|$
$g : \mathbb{Z}_+ \times \mathbb{Z}_+ \to \mathbb{Z}_+$, $g(m, n) = m$ is surjective

Challenge

Problem

Show that $\mathbb{Z}_+$ and $\mathbb{Z}_+ \times \mathbb{Z}_+$ have the same cardinality.

Hint: consider $f : \mathbb{Z}_+ \times \mathbb{Z}_+ \to \mathbb{Z}_+$, $f(m, n) = 2^m 3^n$

Solution

$2^a 3^b = 2^c 3^d$ if and only if $a = c$ and $b = d$ (Fund. Thm. Arith.)
$f(m, n) = 2^m 3^n$ is injective
$|\mathbb{Z}_+ \times \mathbb{Z}_+| \leq |\mathbb{Z}_+|$
$g : \mathbb{Z}_+ \times \mathbb{Z}_+ \to \mathbb{Z}_+$, $g(m, n) = m$ is surjective
$|\mathbb{Z}_+ \times \mathbb{Z}_+| \geq |\mathbb{Z}_+|$

Challenge

Problem

Show that $\mathbb{Z}_+$ and $\mathbb{Z}$ have the same cardinality.

Challenge

Problem

Show that $\mathbb{Z}_+$ and $\mathbb{Z}$ have the same cardinality.

Hint: consider $f : \mathbb{Z}_+ \to \mathbb{Z}$

$$f(n) = \begin{cases} n/2 & n \text{ is even} \\ -(n-1)/2 & n \text{ is odd} \end{cases}$$

Finite and infinite

Finite and infinite

Sets with finite cardinality:

$$\{1, 2, 3\}, \quad \{\mathbb{Z}, \mathbb{R}\}, \quad \{x : x \text{ is a student at CSUF}\}$$

Finite and infinite

Sets with finite cardinality:

$$\{1,2,3\}, \quad \{\mathbb{Z},\mathbb{R}\}, \quad \{x : \ x \text{ is a student at CSUF}\}$$

Sets with infinite cardinality:

$$\mathbb{Z}_+, \quad \mathbb{Q}, \quad \mathbb{R}, \quad \mathbb{C}.$$

Finite and infinite

Sets with finite cardinality:

$$\{1, 2, 3\}, \quad \{\mathbb{Z}, \mathbb{R}\}, \quad \{x : x \text{ is a student at CSUF}\}$$

Sets with infinite cardinality:

$$\mathbb{Z}_+, \quad \mathbb{Q}, \quad \mathbb{R}, \quad \mathbb{C}.$$

Cantor's discovery: there are multiple sizes of infinity!

$$\mathbb{Z}_+,\ \mathbb{Z},\ \mathbb{Z} \times \mathbb{Z},\ \mathbb{Q},\ \text{are all the same cardinality}$$

Finite and infinite

Sets with finite cardinality:

$$\{1,2,3\}, \quad \{\mathbb{Z},\mathbb{R}\}, \quad \{x : \ x \text{ is a student at CSUF}\}$$

Sets with infinite cardinality:

$$\mathbb{Z}_+, \quad \mathbb{Q}, \quad \mathbb{R}, \quad \mathbb{C}.$$

Cantor's discovery: there are multiple sizes of infinity!

$$\mathbb{Z}_+,\ \mathbb{Z},\ \mathbb{Z}\times\mathbb{Z},\ \mathbb{Q},\ \text{are all the same cardinality}$$

$$\mathbb{R} \text{ has larger cardinality than } \mathbb{Z}_+.$$

Cantor diagonalization

Suppose there were a bijection

$$f : \mathbb{Z}_+ \to (0, 1)$$

Cantor diagonalization

Suppose there were a bijection

$$f : \mathbb{Z}_+ \to (0,1)$$

$$\begin{aligned}
f(1) &= 0.a_{11}a_{12}a_{13}a_{14}a_{15}\dots \\
f(2) &= 0.a_{21}a_{22}a_{23}a_{24}a_{25}\dots \\
f(3) &= 0.a_{31}a_{32}a_{33}a_{34}a_{35}\dots \\
f(4) &= 0.a_{41}a_{42}a_{43}a_{44}a_{45}\dots \\
f(5) &= 0.a_{51}a_{52}a_{53}a_{54}a_{55}\dots \\
&\vdots
\end{aligned}$$

Cantor diagonalization

Suppose there were a surjection

$$f : \mathbb{Z}_+ \to (0, 1)$$

Cantor diagonalization

Suppose there were a surjection

$$f : \mathbb{Z}_+ \to (0,1)$$

$$\begin{aligned}
f(1) &= 0.a_{11}a_{12}a_{13}a_{14}a_{15}\dots \\
f(2) &= 0.a_{21}a_{22}a_{23}a_{24}a_{25}\dots \\
f(3) &= 0.a_{31}a_{32}a_{33}a_{34}a_{35}\dots \\
f(4) &= 0.a_{41}a_{42}a_{43}a_{44}a_{45}\dots \\
f(5) &= 0.a_{51}a_{52}a_{53}a_{54}a_{55}\dots \\
&\vdots
\end{aligned}$$

Cantor diagonalization

Now define $b_1, b_2, b_3 \cdots \in \{0, 9\}$ by

$$b_j = \begin{cases} 1, & a_{jj} = 0 \\ 0, & a_{jj} \neq 0 \end{cases}$$

Cantor diagonalization

Now define $b_1, b_2, b_3 \cdots \in \{0, 9\}$ by

$$b_j = \begin{cases} 1, & a_{jj} = 0 \\ 0, & a_{jj} \neq 0 \end{cases}$$

$$b = 0.b_1 b_2 b_3 b_4 b_5 \ldots$$

Cantor diagonalization

- $b \in (0, 1)$

Cantor diagonalization

- $b \in (0, 1)$
- f is surjective

Cantor diagonalization

- $b \in (0, 1)$
- f is surjective
- there exists $n \in \mathbb{Z}_+$ with $f(n) = b$

Cantor diagonalization

- $b \in (0, 1)$
- f is surjective
- there exists $n \in \mathbb{Z}_+$ with $f(n) = b$

$$0.b_1 b_2 b_3 b_4 b_5 \cdots = 0.a_{n1} a_{n2} a_{n3} a_{n4} a_{n5} \ldots$$

Cantor diagonalization

- $b \in (0, 1)$
- f is surjective
- there exists $n \in \mathbb{Z}_+$ with $f(n) = b$

$$0.b_1 b_2 b_3 b_4 b_5 \cdots = 0.a_{n1} a_{n2} a_{n3} a_{n4} a_{n5} \ldots$$

- $b_n = a_{nn}$

Cantor diagonalization

- $b \in (0, 1)$
- f is surjective
- there exists $n \in \mathbb{Z}_+$ with $f(n) = b$

$$0.b_1 b_2 b_3 b_4 b_5 \cdots = 0.a_{n1} a_{n2} a_{n3} a_{n4} a_{n5} \ldots$$

- $b_n = a_{nn}$ CONTRADICTION!

Countable and uncountable

A set is

Countable and uncountable

A set is

- **countably infinite** if it has the same cardinality as $\mathbb{Z}_+$

Countable and uncountable

A set is

- **countably infinite** if it has the same cardinality as $\mathbb{Z}_+$
- **countable** if it is finite or countably infinite

Countable and uncountable

A set is

- **countably infinite** if it has the same cardinality as $\mathbb{Z}_+$
- **countable** if it is finite or countably infinite
- **uncountable** otherwise

Countable and uncountable

A set is

- **countably infinite** if it has the same cardinality as $\mathbb{Z}_+$
- **countable** if it is finite or countably infinite
- **uncountable** otherwise

Theorem (Cantor's Theorem)

$\mathbb{R}$ is uncountable

Outline

Ultimate Cantor diagonalization

The **power set** $\mathcal{P}(A)$ of a set A is

$$\mathcal{P}(A) = \{S : S \subseteq A\}.$$

Ultimate Cantor diagonalization

The **power set** $\mathcal{P}(A)$ of a set A is

$$\mathcal{P}(A) = \{S : S \subseteq A\}.$$

Theorem

The cardinality of $\mathcal{P}(A)$ is strictly larger than A.

Ultimate Cantor diagonalization

The **power set** $\mathcal{P}(A)$ of a set A is

$$\mathcal{P}(A) = \{S : S \subseteq A\}.$$

Theorem

The cardinality of $\mathcal{P}(A)$ is strictly larger than A.

Proof.

Ultimate Cantor diagonalization

The **power set** $\mathcal{P}(A)$ of a set A is

$$\mathcal{P}(A) = \{S : S \subseteq A\}.$$

Theorem

The cardinality of $\mathcal{P}(A)$ is strictly larger than A.

Proof.

Suppose $f : A \to \mathcal{P}(A)$ is surjective.

Ultimate Cantor diagonalization

The **power set** $\mathcal{P}(A)$ of a set A is

$$\mathcal{P}(A) = \{S : S \subseteq A\}.$$

Theorem

The cardinality of $\mathcal{P}(A)$ is strictly larger than A.

Proof.

Suppose $f : A \to \mathcal{P}(A)$ is surjective.
Consider the set

$$S = \{a \in A : a \notin f(a)\}.$$

$\square$

Challenge

Challenge

Since f is surjectve, $S = f(x)$ for some $x \in A$.

Challenge

Since f is surjectve, $S = f(x)$ for some $x \in A$.

Problem

Consider the statement $x \in S$. What can you conclude?

Challenge

Problem

Is there a set with cardinality larger than $\mathbb{R}$?

Challenge

Problem

Show that the cardinality of the line segment $(0, 1)$ and the square $(0, 1) \times (0, 1)$ is the same.

Countable unions

A **countable family of sets** is a family of sets where the index set is countable.

Countable unions

A **countable family of sets** is a family of sets where the index set is countable.

Theorem

The union of a countable family of countable sets is countable.

Countable unions

A **countable family of sets** is a family of sets where the index set is countable.

Theorem

The union of a countable family of countable sets is countable.

Proof.

Let $\{A_i : i \in I\}$ be a countable family of countable sets.

Countable unions

A **countable family of sets** is a family of sets where the index set is countable.

Theorem

The union of a countable family of countable sets is countable.

Proof.

Let $\{A_i : i \in I\}$ be a countable family of countable sets.
I is countable.

Countable unions

A **countable family of sets** is a family of sets where the index set is countable.

Theorem

The union of a countable family of countable sets is countable.

Proof.

Let $\{A_i : i \in I\}$ be a countable family of countable sets.
I is countable. There is a surjection $\mathbb{Z}_+ \to I$

Countable unions

A **countable family of sets** is a family of sets where the index set is countable.

Theorem

The union of a countable family of countable sets is countable.

Proof.

Let $\{A_i : i \in I\}$ be a countable family of countable sets.
I is countable. There is a surjection $\mathbb{Z}_+ \to I$
In other words, we can enumerate $I = \{i_1, i_2, i_3, \dots\}$.

Countable unions

A **countable family of sets** is a family of sets where the index set is countable.

Theorem

The union of a countable family of countable sets is countable.

Proof.

Let $\{A_i : i \in I\}$ be a countable family of countable sets.
I is countable. There is a surjection $\mathbb{Z}_+ \to I$
In other words, we can enumerate $I = \{i_1, i_2, i_3, \dots\}$.
A_i is countable for all $i \in I$.

Countable unions

A **countable family of sets** is a family of sets where the index set is countable.

Theorem

The union of a countable family of countable sets is countable.

Proof.

Let $\{A_i : i \in I\}$ be a countable family of countable sets.
I is countable. There is a surjection $\mathbb{Z}_+ \to I$
In other words, we can enumerate $I = \{i_1, i_2, i_3, \dots\}$.
A_i is countable for all $i \in I$.
So again $A_i = \{a_{i1}, a_{i2}, a_{i3}, \dots\}$.

Countable unions

A **countable family of sets** is a family of sets where the index set is countable.

Theorem

The union of a countable family of countable sets is countable.

Proof.

Let $\{A_i : i \in I\}$ be a countable family of countable sets.
I is countable. There is a surjection $\mathbb{Z}_+ \to I$
In other words, we can enumerate $I = \{i_1, i_2, i_3, \dots\}$.
A_i is countable for all $i \in I$.
So again $A_i = \{a_{i1}, a_{i2}, a_{i3}, \dots\}$.
Define $f : \mathbb{Z}_+ \times \mathbb{Z}_+ \to \bigcup_{i \in I} A_i$

Countable unions

A **countable family of sets** is a family of sets where the index set is countable.

Theorem

The union of a countable family of countable sets is countable.

Proof.

Let $\{A_i : i \in I\}$ be a countable family of countable sets.
I is countable. There is a surjection $\mathbb{Z}_+ \to I$
In other words, we can enumerate $I = \{i_1, i_2, i_3, \dots\}$.
A_i is countable for all $i \in I$.
So again $A_i = \{a_{i1}, a_{i2}, a_{i3}, \dots\}$.
Define $f : \mathbb{Z}_+ \times \mathbb{Z}_+ \to \bigcup_{i \in I} A_i$
$f(j,k) = a_{i_j,k}$.

Countable unions

A **countable family of sets** is a family of sets where the index set is countable.

Theorem

The union of a countable family of countable sets is countable.

Proof.

Let $\{A_i : i \in I\}$ be a countable family of countable sets.
I is countable. There is a surjection $\mathbb{Z}_+ \to I$
In other words, we can enumerate $I = \{i_1, i_2, i_3, \dots\}$.
A_i is countable for all $i \in I$.
So again $A_i = \{a_{i1}, a_{i2}, a_{i3}, \dots\}$.
Define $f : \mathbb{Z}_+ \times \mathbb{Z}_+ \to \bigcup_{i \in I} A_i$
$f(j, k) = a_{i_j,k}$. Surjection! $\square$

Challenge!

A real number is called **algebraic** if it is a root of a polynomial with integer coefficients.

Challenge!

A real number is called **algebraic** if it is a root of a polynomial with integer coefficients.

Problem

Prove that the set of all algebraic numbers is countable.

Challenge!

A real number is called **algebraic** if it is a root of a polynomial with integer coefficients.

Problem

Prove that the set of all algebraic numbers is countable.

Hint: use the previous theorem!